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Effect of Transparency Levels and Real-World Backgrounds on the User Interface in Augmented Reality Environments

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ABSTRACT
With the introduction of the metaverse, augmented reality (AR) is expected to become the next interface between people and computers. Research is conducted to improve the usability of AR applications and devices. Color transparency and background effect are the two critical factors to consider while developing AR applications. Ten user interface conditions (5 color transparency levels × 2 background conditions) were evaluated in an AR environment. The task completion time and mean error were noted along with the subjective scores of button visibility, background visibility, and motion sickness. The results revealed a significant difference in task completion time, button visibility, and background visibility between transparency conditions, but not in the number of errors and total motion sickness. The transparency level of T0.5 and T0.75 is recommended for the interface of AR applications. This study could be helpful when creating augmented reality applications for the metaverse and other platforms.

KEYWORDS
Augmented reality; transparency; visibility; user interface

1. Introduction

Display technologies have successfully progressed from the large cathode ray tube to small flat panel designs, such as liquid crystal display (LCD) and organic light-emitting diode (OLED), since the start of this millennium (Chen et al., 2017). More recently, the next-generation display technologies being developed go beyond flat panels that are only placed in front of users and instead aim to revolutionize how people interact with their surroundings (Cakmakci & Rolland, 2006). One such display is augmented reality, which strives for high-quality see-through performance and enhances the real world by superimposing digital content. As the next generation of interactive displays, augmented reality (AR) and virtual reality (VR) headsets can deliver rich three-dimensional (3D) visual experiences (Cakmakci & Rolland, 2006; Yin et al., 2021; Zhan et al., 2020). Augmented reality will be a revolution in human-computer interaction. Both augmented reality and virtual reality displays can lead to appealing applications in health care, engineering design, education, manufacturing, retail, and entertainment, thanks to their cutting-edge optical technology and novel user experiences (Caudell & Mizell, 1992; Munsinger et al., 2019; Rendon et al., 2012; Wu et al., 2013). AR has been a potent tool for enhancing process flexibility and efficiency in industrial operations (Fraga-Lamas et al., 2018).

There is no particular display technology that is required for AR (van Krevelen & Poelman, 2015). The term "AR display" refers to any transparent display that can instruct users while still allowing them to see the real environment. AR and VR devices are emerging as next-generation interactive displays that can provide vibrant 3D visual experiences replacing computers and smartphones. While VR emphasizes an entirely immersive experience, AR encourages user interaction with digital content and the actual environment (Xiong et al., 2021). Field of view (FoV), eye box, angular resolution, dynamic range, the proper depth cue, and other ergonomic concerns are some of the display-related difficulties that AR and VR share to meet the demanding requirements of human vision (Xiong et al., 2021). These requirements frequently involve trade-offs with one another. Several AR HMDs, such as google glass and Microsoft Hololens, are now available to experience AR environments.

Designing content for augmented and mixed reality requires careful consideration of color, lighting, and materials for all your virtual assets or object (Hamacher et al., 2019). Research has been done on various issues with augmented reality, including interface design, interaction techniques, motion sickness, and other usability issues with augmented reality devices (Aromaa et al., 2020; Datcu et al., 2015; Dey et al., 2018; Gabbard et al., 2008; Hamacher et al., 2019; Muhammad et al., 2021). Several other studies provide insights into the state of the art of AR research (Azuma, 1997; Erickson et al., 2020; Gabbard et al., 2005; Milgram & Kishino, 1994; Satkowski & Dachselt, 2021). Transparency is the quality of being easily see-through. The definition,

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according to the International Lighting Vocabulary by International Commission on Illumination (CIE), is "the degree of visibility of an object through a medium" (CIE, 2022). The color appearance of transparent objects is a significant and challenging problem since it depends on various elements, such as ambient light, the virtual object's color appearance, and the background's color in the actual world. A solid black object will appear no different from the real world. Without taking color transparency into account, color-matching requirements for augmented reality were investigated (Zhang et al., 2018). In augmented reality, transparency differs from simulated or subtractive filter transparency because of additional light overlaid (Zhang et al., 2021). It is essential to study and control the color transparency of content for many AR applications similar to display technology. A transparent display has been mostly used for AR devices, and various studies have been performed to improve its performance technically. Transparency perception in terms of general color contrast and brightness was also studied (Ekroll & Faul, 2013; Murdoch, 2020). There is a lack of user-based studies related to the usability of color transparency and its background in augmented reality environments.

A color consists of four components which are red, green, blue, and alpha. The first three are obvious: red, green, and blue. The fourth component in the RGBA model, known as the alpha (A) component, determines how opaque or transparent a color is. A number of 0 is totally transparent, while a value of 1 is entirely solid (Porter & Duff, 1984). The Alpha channel of color can control the transparency of a virtual object, so the color transparency was controlled by the alpha channel of the color in the current study. To assess the background effect, the transparency conditions were studied in two different scenarios. Previous studies discussed various color blending and color correction models based on background color and object color but they didn't consider the performance due to changes in alpha channel for the same color based on a user study. We analyzed the impact of varying color transparency and background levels according to task completion time, number of errors, button visibility, background visibility, and motion sickness. Thus, an experiment was conducted in an AR environment considering five levels of color transparency and two background conditions (Uniform and Non-uniform). The following hypothesis was driven for the current study. H1: There is no significant difference between the task completion times, the number of errors, motion sickness, and subjective visibility scores of different color transparencies in an AR environment. H2: There is no significant difference between the task completion times, number of errors, motion sickness, and subjective visibility scores of different background conditions in the AR environment.

2. Related work

The influence of color and background texture on visibility and text legibility in an AR environment is widely acknowledged, but only a small amount of research has examined and measured this impact. Several recent studies have explored the effects of these factors on the user experience in AR environments and have found that they play a significant role in determining the overall quality and usability of AR interfaces. Perceptual problems, including color perception, lighting, text legibility, and color blending, are some of the most significant human factors issues in optical see-through AR. Early attempts to create AR graphics in outdoor environments revealed the well-known problem of lighting issues, such as certain colors (e.g., magenta and cyan) appearing translucent and challenging to see due to washout in sunlight (Livingston et al., 2009; Thomas et al., 2002). The quality of AR users' experiences is influenced by optical elements, display technology, and visual context (Livingston et al., 2009). Perceived colors and lighting of AR graphics can become distorted under frequent changes in lighting and real-world backgrounds, leading to poor visual presentation for users (Kruijff et al., 2010). Another issue is AR text legibility, for which it is already established that the colors and textures of the real-world background have an impact on how easy it is to read AR text in various industrial applications of AR. This means that changes in the background condition can cause visual degradation of 3D textures, causing distortion and masking effects on the displayed text (Di Donato et al., 2015). Gabbard et al. have introduced a technique to systematically investigate the legibility of various AR text colors against predetermined backgrounds (Gabbard et al., 2006, 2007). The research conducted by Zhang et al. did not address color compensation directly. Several studies (Hirobe et al., 2022; Zhang et al., 2018, 2021) performed color-matching experiments using an optical see-through (OST) HMD and estimated the reference color perceived by users through the device. To do this, they analyzed the spectral transmittance of both the original reference color and the OST-HMD. The common factor in their background color estimations was their emphasis on the spectral transmittance of the OST-HMD, which they viewed as the primary contributor to shifts in the background color.

Despite various methods used to correct color blending in AR, researchers in this field agree that it remains a challenging task, particularly due to the lack of a one-size-fits-all calibration strategy for different AR display types with varying specifications. Therefore, there is a growing need for more user-based, psychophysical studies that can explore the relationship between physical stimuli such as AR graphics, real-world backgrounds, and lighting, and the resulting perceptions and sensations, such as subjective color judgments and visibility. Such kind of visibility issues for human avatars in an AR environment due to transparency are discussed by Peck et al. (2022). Despite attempts to address this issue using different methods, there is a scarcity of published research that involves actual users to examine the impact of background conditions and color opaqueness or transparency on AR user performance. In this study, the color change and visibility issues of a virtual object along with performance due to change in its alpha value are studied in an AR environment.

3. Method

3.1. Participants

In this study, twenty-five students participated as experimentees with a mean age of 24.5 (± 4.4). The number of participants was chosen based on previous similar studies (Gabbard et al., 2005, 2006, 2022; Murdoch, 2020; Yamin et al., 2023). Twelve of them were female, and thirteen were male. Eighteen subjects had normal vision, compared to seven who experimented with glasses. No participants were excluded based on age or vision. The participants were physically fit. They were provided an incentive of $20 to participate in the experiment.

3.2. Apparatus

The AR device used in this study was Microsoft HoloLens-2 (Microsoft 2022). It includes optical see-through holographic lenses with 2k 3:2 light engine resolution. The field of view of HoloLens-2 was 3° 50″. The holographic density was greater than 2.5k radiant (light points per radian). Hand tracking was used to interact with HoloLens, and the prototype was created in Unity 3D using C# scripting. A seminar room is used as an experimental room for this study with a controlled lighting condition of 790–800 lux level during daylight hours (Figure 1). The lighting in the experimental room was measured using a lux meter and kept the same for all experimental conditions. The brightness levels were monitored during the experiment, and the HoloLens 2 Level 5 setting was applied. The black screen was used to control the background color effect during the experiment for uniform conditions. The prototype consists of nine square buttons (3×3 array). The distance between the virtual object and the eye was 120 cm, and the field of view of the buttons was 3° 50″ (Figure 2). The color model considered for the prototype was RGBA with R: 28, G: 207, and B: 30, whereas five conditions according to the value of the alpha channel (0, 0.25, 0.5, 0.75, and 1) were considered to analyze the color transparency. The color code in terms of HSV space was H: 121°, S: 86.5%, and V:81.2%. A color with an alpha value of 0 is completely transparent, whereas a value of 1 is solid (opaque) (Kia et al., 2021).

3.3. Measure

In this study, the conditions are referred to as T0, T0.25, T0.5, T0.75, and T1 for the alpha value of 0, 0.25, 0.5, 0.75, and 1, respectively in RGBA. Two background conditions were studied to analyze the background effect as well. The dependent variables in this study were the task completion time, the number of errors, and subjective scores, whereas the independent variables were the transparency level (Five levels; T0, T0.25, T0.5, T0.75, T1), and background condition (Two levels; Uniform and Non-uniform).

Figure 1. Experimental environment.

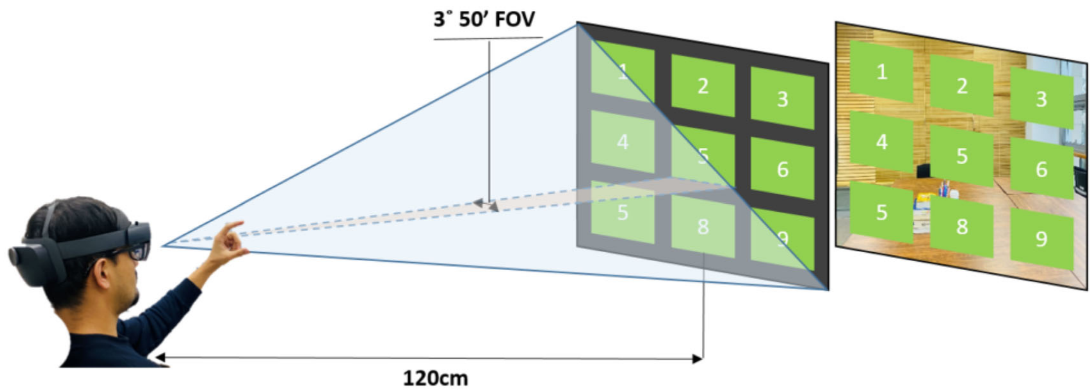

Figure 2. Prototype model with uniform and non-uniform background.

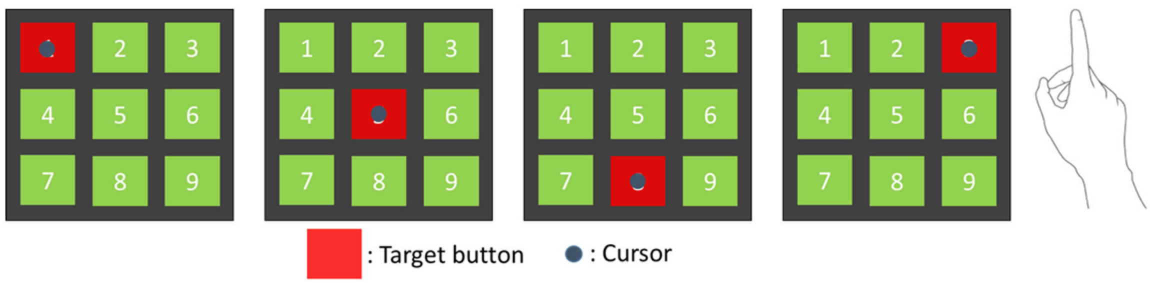

Figure 3. A button selection task set with interaction type.

3.4. Tasks

The total number of experimental conditions was ten (5 transparencies × 2 backgrounds). Four-button selections were repeated five times for each treatment condition (4 button selections × 5 repetitions × 10 experimental conditions), and each participant performed 200 button selections in the entire experiment (Figure 3). The experiment was set up to be within-subject. The task was to select a randomly highlighted button and rate the subjective questionnaire for background visibility, button visibility, and motion sickness.

3.5. Procedure

The participants were informed of the study's goal before the experiment, and their written agreement was acquired. Additionally, they were free to exit the experiment whenever they pleased. The participants provided their personal information by completing a form and submitting the necessary paperwork. They were given instructions on how to use HoloLens and how it interacts with hand motions because some of them were unfamiliar with it. A practice session was held through the demonstration task to help the attendees become more comfortable using HoloLens. Each participant had their HoloLens calibrated using the default method. The experiment was conducted with a 2-minute rest period between the ten experimental conditions (5 transparency levels and 2 backgrounds) (Figure 4). After completing the experiment under each condition, the participants were asked to complete a questionnaire for a subjective assessment of button visibility and background visibility. The questions used to get subjective feedback regarding background and button visibility were: (1) Was the background easily visible with this transparency? (2) Was the button easily visible with this transparency?, respectively. The 5-point Likert-type scale from 1 to 5 (1 = Extremely Not, 2 = Almost Not, 3 = Average, 4 = Almost Yes, 5 = Extremely Yes) was used to collect the subjective scores. After completing each set of tasks, they also completed the Simulator Sickness Questionnaire (SSQ) based on each condition. The learning effects were avoided by using a balanced Latin square pattern. Each participant spent about 90 minutes on the entire experiment.

3.6. Analysis

The data were analyzed using univariate analysis of variance (ANOVA) with the task completion time, error rate, and satisfaction score as the dependent variables. Two levels of background condition and five levels of transparency were regarded as fixed parameters. In the statistical models, the individual differences among the subjects were considered random factors. All analyses were carried out using IBM SPSS Statistics 25, and the statistical significance was set at $p = 0.05$. Univariate ANOVA was used to examine the main effect and interaction effects across all treatment conditions, and the Student–Newman-Keuls test was used for post hoc analysis. Analyzing the number of errors, visibility scores, and simulator sickness scores all used the same methodology.

Figure 4. A pictorial display of ten conditions according to transparency and background.

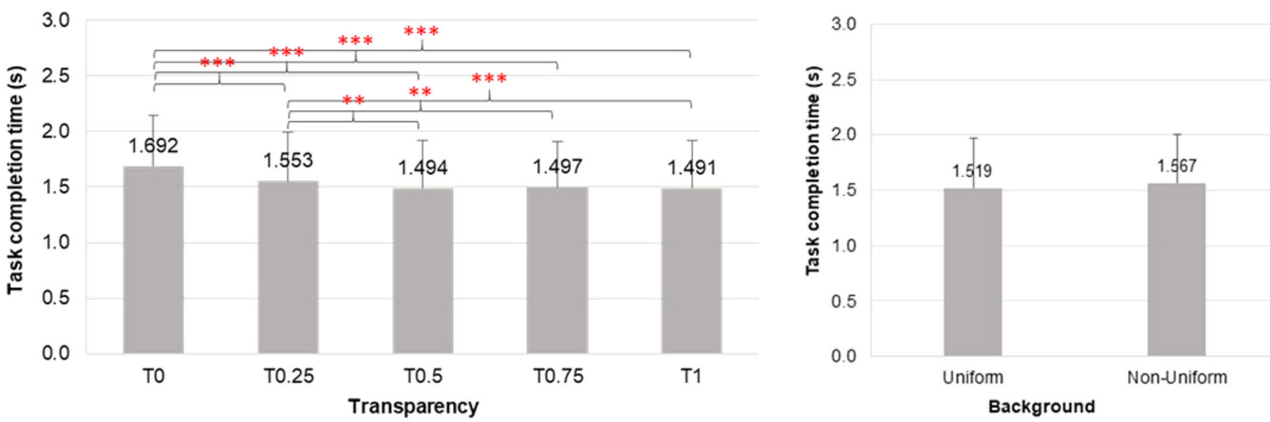

Figure 5. Task completion time according to transparency and background conditions (**$p < 0.01$, ***$p < 0.001$).

4. Results

4.1. Task completion time

According to the statistical results, there is a significant difference in task completion time among the color transparency levels ($F = 17.273$, $p = 0.001$, $\eta_p^2 = 0.418$), but no significant difference among the background conditions ($F = 0.678$, $p = 0.419$, $\eta_p^2 = 0.027$). The post hoc test for transparency level revealed a significant difference in task completion time among T0, T0.25, T0.5, T0.75, and T1 (Figure 5). There is no statistically significant interaction between transparency level and background condition for task completion time ($F = 1.044$, $p = 0.389$, $\eta_p^2 = 0.42$).

4.2. Number of errors

For the number of errors, the results show that there is no significant difference among both color transparency levels ($F = 0.651$, $p = 0.628$, $\eta_p^2 = 0.026$) and background conditions ($F = 0.605$, $p = 0.444$, $\eta_p^2 = 0.24$). There is also no interaction effect between transparency levels and background conditions for the number of errors ($F = 0.230$, $p = 0.921$, $\eta_p^2 = 0.009$) (Figure 6).

4.3. Visibility

4.3.1. Button visibility

The results of subjective scores show that there is a significant difference in the scores of button visibility among both color transparency levels ($F = 32.895$, $p = 0.001$, $\eta_p^2 = 0.578$) and background conditions ($F = 27.828$, $p = 0.001$, $\eta_p^2 = 0.537$). According to the *post-hoc* result, there is a significant difference between button visibility scores of T0, T0.25, T0.5, T0.75, and T1 (Figure 7). For the interaction effect, there is a statistically significant interaction between transparency levels and background conditions for button visibility in augmented reality ($F = 7.108$, $p = 0.001$, $\eta_p^2 = 0.228$) (Figure 8).

4.3.2. Background visibility

The results of subjective scores show that there is a significant difference among the scores of background visibility of color transparency levels ($F = 10.92$, $p = 0.001$, $\eta_p^2 = 0.312$) but not among the background conditions ($F = 0.993$, $p = 0.329$, $\eta_p^2 = 0.040$). The post-hoc results revealed a significant difference between the background visibility scores of T0, T0.25, T0.5, T0.75, and T1 (Figure 9). For the interaction effect, there is no statistically significant interaction

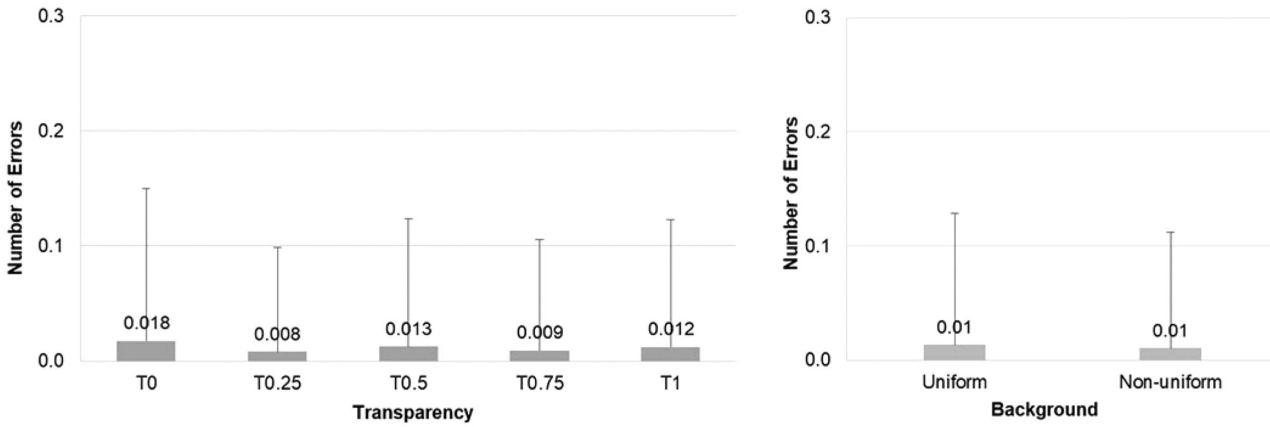

Figure 6. The number of errors according to transparency and background conditions.

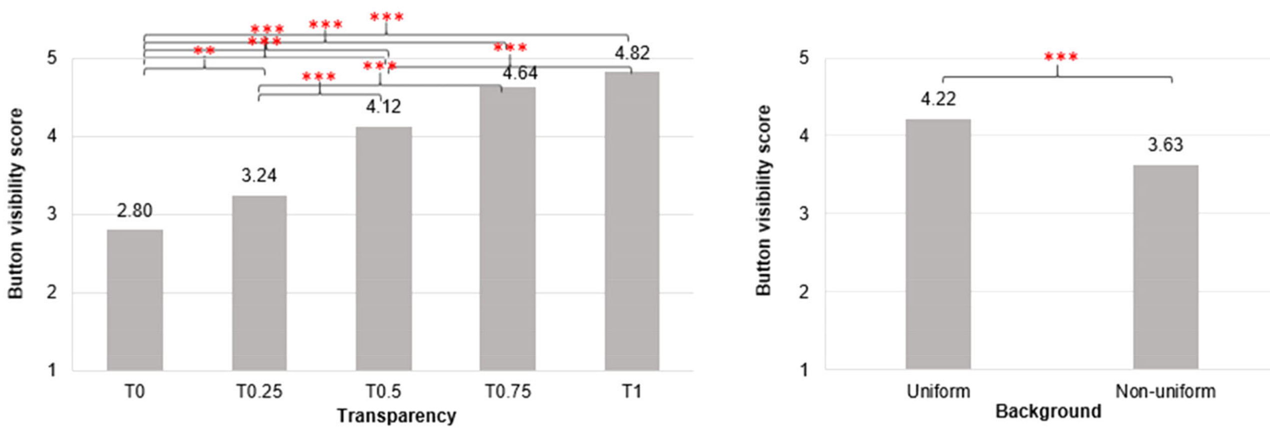

Figure 7. Button visibility scores according to transparency and background conditions (**$p < 0.01$, ***$p < 0.001$).

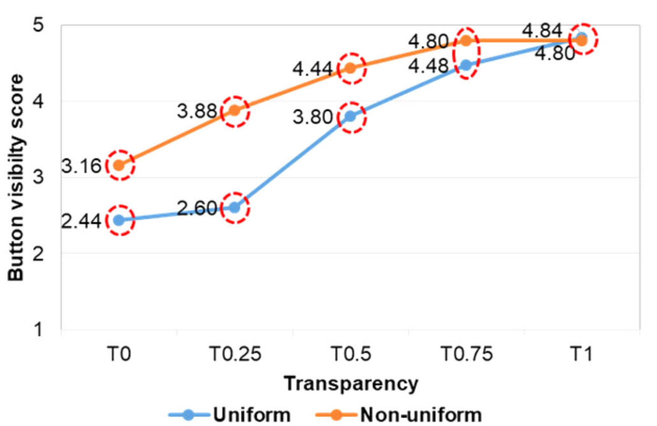

Figure 8. Interaction effect between transparency and background for button visibility. Different circles represent a significant difference.

between transparency levels and background conditions for background visibility in augmented reality ($F = 2.239$, $p = 0.070$, $\eta_p^2 = 0.085$).

4.4. Motion sickness

According to the motion sickness results, there is no significant difference in nausea ($F = 1.847$, $p = 0.126$, $\eta_p^2 = 0.071$) ($F = 1.189$, $p = 0.286$, $\eta_p^2 = 0.047$), oculomotor ($F = 1.401$, $p = 0.240$, $\eta_p^2 = 0.055$) ($F = 2.857$, $p = 0.104$, $\eta_p^2 = 0.106$) and total sickness ($F = 1.168$, $p = 0.330$, $\eta_p^2 = 0.046$) ($F = 3.957$, $p = 0.058$, $\eta_p^2 = 0.142$) among transparency levels and background conditions, respectively, but a significant difference in disorientation is observed among both transparency levels ($F = 3.314$, $p = 0.014$, $\eta_p^2 = 0.121$) and background conditions ($F = 4.541$, $p = 0.044$, $\eta_p^2 = 0.159$). *Post-hoc* results revealed that the transparency level of T0.25 has significantly more disorientation than T0.75, and T1 (Figure 10, Table 1).

5. Discussion

5.1. Transparency levels

This study analyzed the five transparency levels according to task completion time, mean error, button visibility, background visibility, and motion sickness. Alpha compositing, also known as alpha blending, is the act of integrating a single picture with a background to provide the impression of partial or complete transparency in computer graphics (PCMAG, 2022). Compositing is used extensively when combining computer-rendered images with real-world or live footage. This study's results revealed a significant main effect between transparency levels regarding task completion time, button visibility, and background visibility. T0 and T1 took the most and least task completion time, respectively. T0 is perceived as transparent, and T1 is perceived to be the

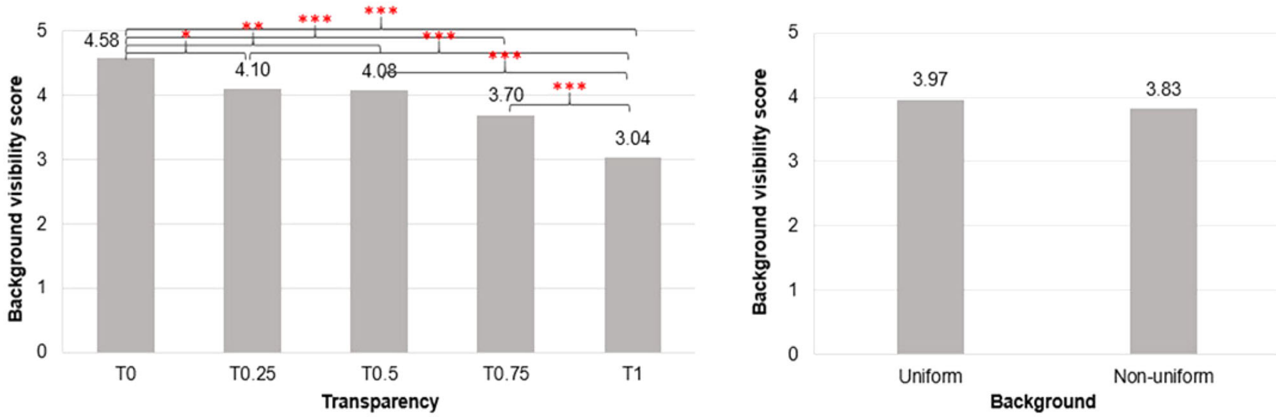

Figure 9. Background visibility scores according to transparency and background conditions (* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$).

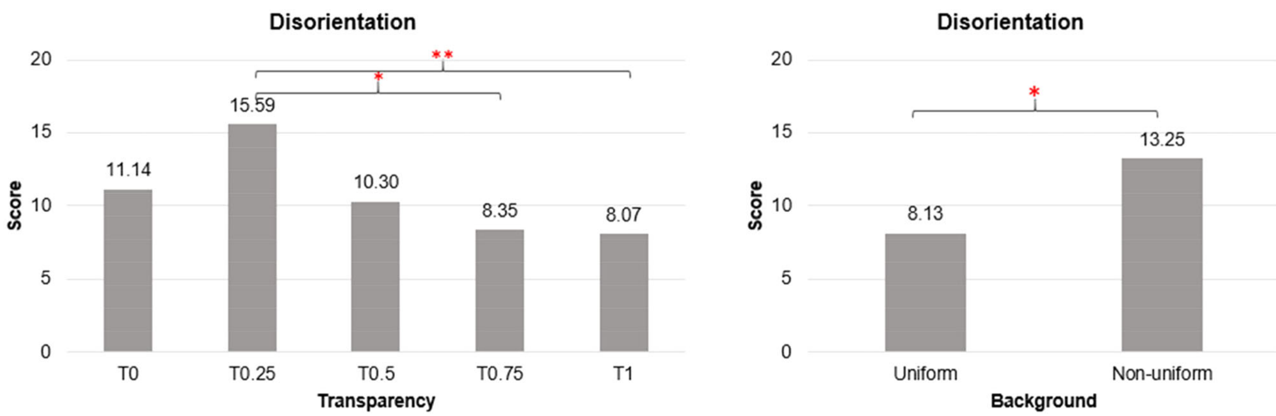

Figure 10. Disorientation scores according to transparency and background conditions (* $p < 0.05$; ** $p < 0.01$).

most opaque of the tested conditions, so we will focus on other levels for the AR environment. According to the study by Zhang et al. (2021), with a solid background, the AR object will be perceived to be completely opaque regardless of the luminance of the additional AR overlay. We considered T0 and T1 in this study to analyze the trend for future studies concerning AR transparency. Among others, T0.5 has the lowest task completion time, followed by T0.75 and T0.25. No significant main effect in mean error and motion sickness was observed between the transparency levels. The button visibility score for T0.5, T0.75, and T1 was more than four, which is interpreted as "Almost Yes." T0.5 and T0.75 are most favorable regarding button visibility in AR environments. Off course, the button visibility at T1 is highest with both uniform and non-uniform backgrounds due to its characteristics of opacity. We excluded it because of the need for an AR environment where we can easily see both object and background. With the factors like task completion time and button visibility, it is vital to consider the background visibility as the study is based on AR, where users have to see both the virtual object and the real world. To balance the trade-off between the visibility of virtual objects and the background, we analyzed the background visibility during task performance. According to the results, there was a significant difference between the background visibility scores of transparency levels. The background visibility score for T0, T0.25, T0.5, T0.75, and T1 are 4.58, 4.10, 4.08, 3.70, and 3.04, respectively. For motion sickness, there was no significant main effect between transparency levels for total sickness, nausea, and oculomotor, but a significant difference was found for disorientation. The non-uniform background was significantly affecting the disorientation more than the uniform background. The recommended transparency levels according to the results of background visibility are T0.25 and T0.5, but the disorientation score was high for T0.25 and T0. We can say that the level of T0.25 is more demanding in terms of focusing due to low visibility. Participants were feeling blurred in this transparency level, which might have contributed more to the disorientation.

5.2. Background condition

This study also analyzed the background effect on transparency by controlling the lighting condition and room environment. Two types of backgrounds, uniform, and non-uniform background, were studied in a real environment. According to the results of the current study, there is no significant main effect between background conditions in task completion time, mean error, background visibility, and total motion sickness. Still, a significant difference is observed in button visibility score and disorientation. Due to the uniform background, it is more visible than a non-uniform background (Figure 9). We can interpret this result for the AR experiments in a controlled environment that using real black uniform background will not make the scenario a virtual reality scenario, and researchers can conduct their experiments in a controlled background. The results will not be significantly different from the non-uniform background. Motion sickness, except disorientation, is also not significantly different among background conditions.

An interaction effect was observed between the button visibility score of transparency and background condition. There was a significant difference between uniform and non-uniform background at transparency levels of T0, T0.25, and T0.5, but no significant difference was observed between background conditions at transparency levels of T0.75 and T1 (Figure 8).

5.3. Recommendations

According to the results of task completion time, mean error, button visibility, background visibility, and motion sickness, the recommended values of color transparency are T0.5 ($\alpha = 0.5$) and T0.75 ($\alpha = 0.75$) (Figure 11). The optimal value can be between these values, so further study is recommended according to more diverse conditions. According to the results of this study, there are some recommendations for AR researchers and practitioners. The recommended color transparency levels are at $\alpha = 0.5$ and $\alpha = 0.75$. No significant difference between task completion time, mean error, background visibility, and total motion sickness of uniform background and non-uniform background. The button looked more visible with the uniform background than with the non-uniform background.

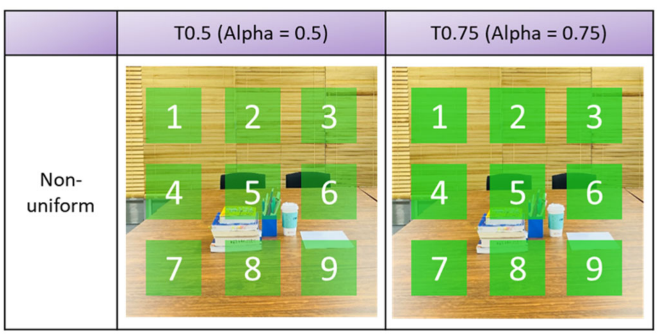

Figure 11. Recommended conditions for AR applications.

5.4. Limitations and future work

This study was conducted in a controlled environment where the light intensity was 790–800 lux. The primary color considered according to the color model was R: 28, G: 207, and B: 30. Such a study can also be conducted for different colors at different lighting conditions according to the working application. The transparency levels considered in this study were only five, so the most optimal condition can be between the recommended values, which is why further investigation in the future is recommended by categorizing the levels further. Studies in more diverse dynamic backgrounds along with other AR devices may give different results, so the authors will consider conducting more indoor and outdoor experiments with different colors and backgrounds. Additionally, we have tried to consider all the AR platforms but there can be a limitation based on the conditions or environment. The suitability of our recommendation may also depend on the specific criteria of the application and user needs. For example, if the AR application is designed for outdoor use in bright sunlight, a higher transparency level may be necessary to ensure the visibility of the interface. Similarly, if the AR application is designed for use by people with visual impairments, a lower transparency level may be necessary to ensure the visibility of the interface. It is important to conduct further research and usability testing for a wide range of AR applications, platforms, and user groups to establish a general guideline for transparency level in AR interfaces.

6. Conclusion

The usability study of the user interface in terms of color transparency levels and background conditions was conducted. Ten interface conditions were analyzed, including five transparency levels and two background conditions (5×2). The task completion time, mean error, button visibility, background visibility, and motion sickness were measured and analyzed. According to the results, there is a

Table 1. Summary of ANOVA main effects.

Dependent variables	Factors	*df*	Mean square	*F*	*p*	η_p^2
Task completion time	Transparency (5 levels)	4	8.047	17.273	0.001***	0.418
	Background condition (2 levels)	1	1.649	0.678	0.419	0.027
	Background condition × Transparency	4	0.390	1.044	0.389	0.042
Mean error	Transparency (5 levels)	4	0.016	0.651	0.628	0.026
	Background condition (2 levels)	1	0.009	0.605	0.444	0.024
	Background condition × Transparency	4	0.004	0.230	0.921	0.009
Button visibility	Transparency (5 levels)	4	38.564	32.895	0.001***	0.578
	Background condition (2 levels)	1	21.316	27.828	0.001***	0.537
	Background condition × Transparency	4	3.016	7.108	0.001***	0.228
Background visibility	Transparency (5 levels)	4	16.430	10.902	0.001***	0.312
	Background condition (2 levels)	1	1.156	0.993	0.329	0.040
	Background condition × Transparency	4	1.246	2.239	0.070	0.085
Total Motion sickness	Transparency (5 levels)	4	2.916	1.168	0.330	0.046
	Background condition (2 levels)	1	0.351	3.957	0.058	0.142
	Background condition × Transparency	4	0.415	1.667	0.164	0.065

***$p < 0.001$

significant difference between task completion time, button visibility, and background visibility of transparency levels. Still, no significant difference was found between mean error and motion sickness. For background conditions, a significant difference is observed in button visibility score but not between task completion times, mean error, and background visibility. Based on the results, the recommended level of color transparency for AR applications is T0.5 and T0.75. This study can be referred to during designing augmented reality applications for metaverse and other AR-related projects.

Disclosure statement

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References

Aromaa, S., Väätänen, A., Aaltonen, I., Goriachev, V., Helin, K., & Karjalainen, J. (2020). Awareness of the real-world environment when using augmented reality head-mounted display. *Applied Ergonomics*, *88*(October), 103145. https://doi.org/10.1016/j.apergo.2020.103145

Azuma, R. T. (1997). A survey of augmented reality. *Presence: Teleoperators and Virtual Environments*, *6*(4), 355–385. https://doi.org/10.1162/pres.1997.6.4.355

Cakmakci, O., & Rolland, J. (2006). Head-worn displays: A review. *Journal of Display Technology*, *2*(3), 199–216. https://doi.org/10.1109/JDT.2006.879846

Caudell, T. P., & Mizell, D. W. (1992). *Augmented reality: An application of heads-up display technology to manual manufacturing processes* [Paper presentation]. Proceedings of the Twenty-Fifth Hawaii International Conference on System Sciences. IEEE. https://doi.org/10.1109/hicss.1992.183317

Chen, H.-W., Lee, J.-H., Lin, B.-Y., Chen, S., & Wu, S.-T. (2017). Liquid crystal display and organic light-emitting diode display: Present status and future perspectives. *Light: Science & Applications*, *7*(3), 17168–17168. https://doi.org/10.1038/lsa.2017.168

CIE. (2022). CIE | International Commission on Illumination/Comission internationale de l'Eclairage/Internationale Beleuchtungskommission. International Commission on Illumination. https://cie.co.at/

Datcu, D., Lukosch, S., & Brazier, F. (2015). On the usability and effectiveness of different interaction types in augmented reality. *International Journal of Human-Computer Interaction*, *31*(3), 193–209. https://doi.org/10.1080/10447318.2014.994193

Dey, A., Billinghurst, M., Lindeman, R. W., & Swan, J. E. (2018). A systematic review of 10 years of augmented reality usability studies: 2005 to 2014. *Frontiers in Robotics and AI*, *5*, 37. https://doi.org/10.3389/frobt.2018.00037

Di Donato, M., Fiorentino, M., Uva, A. E., Gattullo, M., & Monno, G. (2015, June). Text legibility for projected Augmented Reality on industrial workbenches. *Computers in Industry*, *70*, 70–78. https://doi.org/10.1016/j.compind.2015.02.008

Ekroll, V., & Faul, F. (2013). Transparency perception: The key to understanding simultaneous color contrast. *Journal of the Optical Society of America. A, Optics, Image Science, and Vision*, *30*(3), 342–352. https://doi.org/10.1364/JOSAA.30.000342

Erickson, A., Kim, K., Bruder, G., & Welch, G. F. (2020). *A review of visual perception research in optical see-through augmented reality*. The Eurographics Association. https://doi.org/10.2312/egve.20201256

Fraga-Lamas, P., Fernandez-Carames, T. M., Blanco-Novoa, O., & Vilar-Montesinos, M. A. (2018). A review on industrial augmented reality systems for the Industry 4.0 Shipyard. *IEEE Access*, *6*, 13358–13375. https://doi.org/10.1109/ACCESS.2018.2808326

Gabbard, J. L., & Swan, J. E. II. (2008). Usability engineering for augmented reality: Employing user-based studies to inform design. *IEEE Transactions on Visualization and Computer Graphics*, *14*(3), 513–525. https://doi.org/10.1109/TVCG.2008.24

Gabbard, J. L., Smith, M., Merenda, C., Burnett, G., & Large, D. R. (2022). A perceptual color-matching method for examining color blending in augmented reality head-up display graphics. *IEEE Transactions on Visualization and Computer Graphics*, *28*(8), 2834–2851. https://doi.org/10.1109/TVCG.2020.3044715

Gabbard, J. L., Swan, J. E., & Hix, D. (2006). The effects of text drawing styles, background textures, and natural lighting on text legibility in outdoor augmented reality. *Presence: Teleoperators and Virtual Environments*, *15*(1), 16–32. https://doi.org/10.1162/pres.2006.15.1.16

Gabbard, J. L., Swan, J. E., Hix, D., Kim, S.-J., & Fitch, G. (2007). Active *text drawing styles for outdoor augmented reality*: A *user-based study and design implications* [Paper presentation]. 2007 IEEE Virtual Reality Conference (pp. 35–42).

Gabbard, J. L., Swan, J. E., Hix, D., Schulman, R. S., Lucas, J., & Gupta, D. (2005, March). *An empirical user-based study of text drawing styles and outdoor background textures for augmented reality* [Paper presentation]. IEEE Proceedings. VR 2005. Virtual Reality, 2005 (pp. 11–18). IEEE.

Hamacher, A., Hafeez, J., Csizmazia, R., & Whangbo, T. K. (2019). Augmented reality user interface evaluation – performance measurement of Hololens, Moverio and Mouse input. *International Journal of Interactive Mobile Technologies*, *13*(03), 95. https://doi.org/10.3991/ijim.v13i03.10226

Hirobe, D., Uranishi, Y., Orlosky, J., Shirai, S., Ratsamee, P., & Takemura, H. (2022). Characteristics of *background color shifts caused by optical see-through head-mounted displays*. 9 pages. https://doi.org/10.2312/EGVE.20221285

Kia, K., Hwang, J., Kim, I.-S., Ishak, H., & Kim, J. H. (2021). The effects of target size and error rate on the cognitive demand and stress during augmented reality interactions. *Applied Ergonomics*, *97*, 103502. https://doi.org/10.1016/j.apergo.2021.103502

Kruijff, E., Swan, J. E., & Feiner, S. (2010, October). *Perceptual issues in augmented reality revisited* [Paper presentation]. 2010 IEEE International Symposium on Mixed and Augmented Reality (pp. 3–12). IEEE. https://doi.org/10.1109/ISMAR.2010.5643530

Livingston, M. A., Barrow, J. H., & Sibley, C. M. (2009, March). Quantification of *contrast sensitivity and color perception using head-worn augmented reality displays* [Paper presentation]. 2009 IEEE Virtual Reality Conference (pp. 115–122). IEEE. https://doi.org/10.1109/VR.2009.4811009

Microsoft. (2022). HoloLens 2| *Microsoft HoloLens*. Microsoft HoloLens. https://www.microsoft.com/en-us/hololens/buy.

Milgram, P., & Kishino, F. (1994, December). A *taxonomy of mixed reality visual displays*. https://search.ieice.org/bin/summary.php?id=e77-d_12_1321.

Muhammad, H., Jaehyun, P., Yuryeon, L., & Sunyoung, P., K K. H. (2021). Motion sickness indexes in augmented reality environment. *ICIC Express Letters, Part B: Applications*, *12*(12), 1155. https://doi.org/10.24507/icicelb.12.12.1155

Munsinger, B., White, G., & Quarles, J. (2019, September). The *usability of the* Microsoft HoloLens for an *augmented reality game to teach elementary school children* [Paper presentation]. 2019 11th International Conference on Virtual Worlds and Games for Serious Applications (VS-Games) (pp. 1–4). IEEE. https://doi.org/10.1109/VS-Games.2019.8864548

Murdoch, M. J. (2020). Brightness matching in optical see-through augmented reality. *Journal of the Optical Society of America. A, Optics, Image Science, and Vision*, *37*(12), 1927–1936. https://doi.org/10.1364/JOSAA.398931

PCMAG. (2022). Definition of alpha blending. PCMAG. https://www.pcmag.com/encyclopedia/term/alpha-blending.

Peck, T. C., Good, J. J., Erickson, A., Bynum, I., & Bruder, G. (2022). Effects of transparency on perceived humanness: Implications for rendering skin tones using optical see-through displays. *IEEE Transactions on Visualization and Computer Graphics*, *28*(5), 2179–2189. https://doi.org/10.1109/TVCG.2022.3150521

Porter, T., & Duff, T. (1984, January). *Compositing digital images* [Paper presentation]. Proceedings of the 11th Annual Conference on Computer Graphics and Interactive Techniques (pp. 253–259). https://doi.org/10.1145/800031.808606

Rendon, A. A., Lohman, E. B., Thorpe, D., Johnson, E. G., Medina, E., & Bradley, B. (2012). The effect of virtual reality gaming on dynamic balance in older adults. *Age and Ageing*, *41*(4), 549–552. https://doi.org/10.1093/ageing/afs053

Satkowski, M., & Dachselt, R. (2021, May). *Investigating the impact of real-world environments on the perception* of 2D *visualizations in augmented reality* [Paper presentation]. Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems (CHI '21) (Vol. 522, pp. 1–15). Association for Computing Machinery. https://doi.org/10.1145/3411764.3445330

Thomas, B., Close, B., Donoghue, J., Squires, J., Bondi, P. D., & Piekarski, W. (2002). First person indoor/outdoor augmented reality application: ARQuake. *Personal and Ubiquitous Computing*, *6*(1), 75–86. https://doi.org/10.1007/s007790200007

van Krevelen, D. W. F., & Poelman, R. (2010). A survey of augmented reality technologies, applications and limitations. *International Journal of Virtual Reality*, *9*(2), 1–20. https://doi.org/10.20870/IJVR.2010.9.2.2767

Wu, H.-K., Lee, S. W.-Y., Chang, H.-Y., & Liang, J.-C. (2013). Current status, opportunities and challenges of augmented reality in education. *Computers & Education*, *62*, 41–49. https://doi.org/10.1016/j.compedu.2012.10.024

Xiong, J., Hsiang, E.-L., He, Z., Zhan, T., & Wu, S.-T. (2021). Augmented reality and virtual reality displays: Emerging technologies and future perspectives. *Light: Science & Applications*, *10*(1), 216. https://doi.org/10.1038/s41377-021-00658-8

Yamin, P. A. R., Park, J., Kim, H. K., & Hussain, M. (2023). Effects of button colour and background on augmented reality interfaces. *Behaviour & Information Technology*, *0*(0), 1–14. https://doi.org/10.1080/0144929X.2023.2183060

Yin, K., He, Z., Xiong, J., Zou, J., Li, K., & Wu, S.-T. (2021). Virtual reality and augmented reality displays: Advances and future perspectives. *Journal of Physics: Photonics*, *3*(2), 022010. https://doi.org/10.1088/2515-7647/abf02e

Zhan, T., Yin, K., Xiong, J., He, Z., & Wu, S.-T. (2020). Augmented reality and virtual reality displays: Perspectives and challenges. *iScience*, *23*(8), 101397. https://doi.org/10.1016/j.isci.2020.101397

Zhang, L., & Murdoch, M. J. (2021, October). *Perceived transparency in optical see-through augmented reality* [Paper presentation]. 2021 IEEE International Symposium on Mixed and Augmented Reality Adjunct (ISMAR-Adjunct) (pp. 115–120). IEEE. https://doi.org/10.1109/ISMAR-Adjunct54149.2021.00033

Zhang, L., Murdoch, M. J., & Bachy, R. (2021). Color appearance shift in augmented reality metameric matching. *Journal of the Optical Society of America. A, Optics, Image Science, and Vision*, *38*(5), 701–710. https://doi.org/10.1364/JOSAA.420395

Zhang, L., Murdoch, M. J., & Murdoch, M. J. (2018). Color matching criteria in augmented reality. *Journal of Perceptual Imaging*, *1*(1), 10506-1–10506-8. https://doi.org/10.2352/J.Percept.Imaging.2018.1.1.010506

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